

UBER RIDE ANALYSIS

Project Overview:

The objective of this project is to analyze uber ride data to identify travel patterns, peak demand periods, ride duration, and customer behavior. Data visualization and exploratory data analysis (EDA) techniques were used to uncover meaningful insights from the dataset.

Dataset Description:

START_DATE - Date and time when the trip started.

END_DATE - Date and time when the trip ended.

CATEGORY - Type of ride(Business/Personal).

START - Starting location.

STOP - Destination location.

MILES - Distance travelled during the trip.

PURPOSE - Purpose of the trip.

HOUR - Peak hour analysis.

MONTH - Monthly trend analysis

WEEKDAY - Weekly demand analysis.

RIDE_DURATION - Duration calculation.

DAY-NIGHT - Morning/Afternoon/Evening/Night.

ROUND_TRIP - Identify return trips.

DATA PREPROCESSING:

1. Missing Values

The final row contained missing values and was removed.

2. Handling Missing Values

Missing values in the Purpose column were replaced.

3. Standardize Column Names

Changing all the column name to lower cases and remove any characteristics present in the column names.

4. Convert Date Columns

Converting 'start_date', 'end_date' to datetime so that it'll be easy for extraction like hour, month, weekday, ride_duration, day-night.

5. Exploratory Data Analysis

1. Ride Category Distribution

1.1 Visualization

count plot of business vs personal rides.

1.2 Findings

- Business rides dominates the dataset.
- Personal rides account for a much smaller proportion

1.3 Insight

Uber is heavily used for professional travel and meeting.

2. Distribution of Ride Distance

2.1 Visualization

Histogram of miles.

2.2 Findings

- Most trips are short-distance rides.
- Few rides cover long distance.
- Distance distribution is right-skewed.

2.3 Insights

Short trips contribute most to ride volume.

3. Top Pickup Location

3.1 Visualization

The top 10 starting locations.

3.2 Findings

Several locations repeatedly appear as ride origins.

3.3 Insight

These locations represent high-demand pickup zones.

4. Purpose Based Analysis

4.1 Visualization

Count plot of ride purposes.

4.2 Findings

Most common purposes

- Meetings.

- Customer visits.
- Meals/Entertainment.

4.3 Insight

Uber plays an important role in business travel.

5. Month-Wise Ride Analysis

5.1 Visualization

Monthly count plot.

5.2 Findings

Ride demand varies throughout the year.

5.3 Insight

Seasonal demand forecasting can improve driver allocation.

6. Weekday Analysis

6.1 Visualization

Count plot of weekdays.

6.2 Findings

- Weekdays have more rides than weekends.
- Monday-Friday dominate overall ride demand.

6.3 Insights

Demand is strongly linked to office and business schedules.

7. Peak Hour Analysis

7.1 Visualization

Hourly ride count plot

7.2 Findings

Peak demand occurs plot

- Morning – 7 AM – 10 AM
- Evening - 4 PM – 7PM

7.3 Insights

These periods represent commuting hours and require more deliver availability.

8. Day-Night Analysis

8.1 Visualization

Category vs Day-Night count plots.

8.2 Findings

Most rides occur during:

- Morning
- Afternoon

8.2 Insights

Driver scheduling should prioritize daytime hours.

Summary Statistics

Minimum Distance - very short local trips.

Average Distance - Moderate business travel .

Maximum Distance - Long intercity trips

Results indicate:

- Most rides are relatively short.
- Long-duration trips are uncommon.

Key Insights and Trends

Ride Demand Trends

- Business rides dominates the dataset.
- Demand is highest during weekdays.
- Ride volume increases during office commute times.

Peak Hours

Highest ride demand:

- Morning (7-10 AM)
- Evening (4-7 PM)

Lowest ride demand:

- Late night hours

Distance Trends:

- Majority of rides are below average distance.
- Long-distance rides occur infrequently.

Purpose Trends

Top rides purposes include:

1. Meeting
2. Customer visit
3. Meal/Entertainment

These account for a significant share of total trips.

Monthly Trends

Ride demand fluctuates throughout the year

Possible reasons:

- Seasonal travel
- Business cycles
- Holidays

Challenges Faced

Missing Values

Some ride purposes were missing.

Date-Time Formatting

Dates were sorted as text.

Outliers

Some rides had unusually large distances.

Business Application

Driver Allocation

Predict peak demand areas and deploy drivers accordingly.

Customer Experience

Reduce:

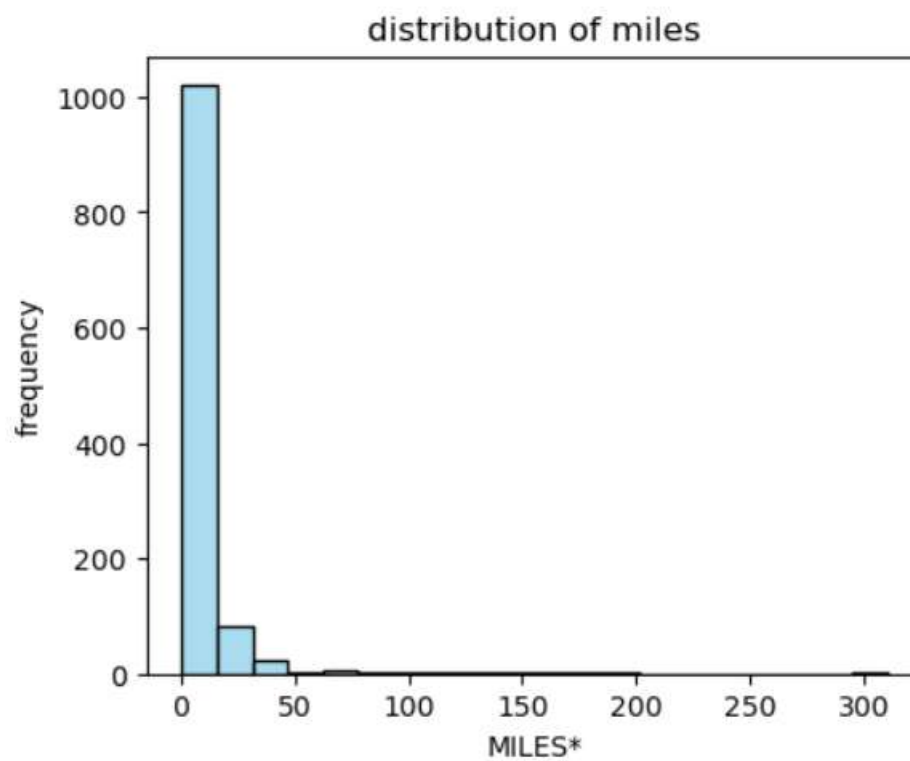
- Waiting time
- Ride cancellations.

Improve:

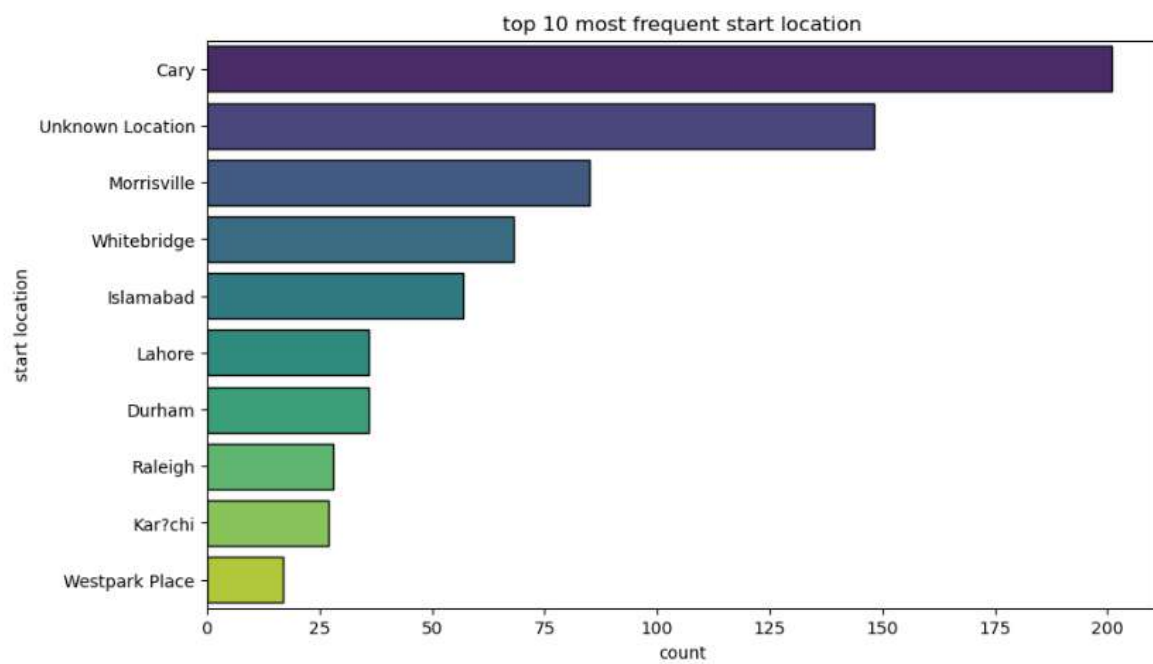
- Availability
- Service quality

RESULTS:

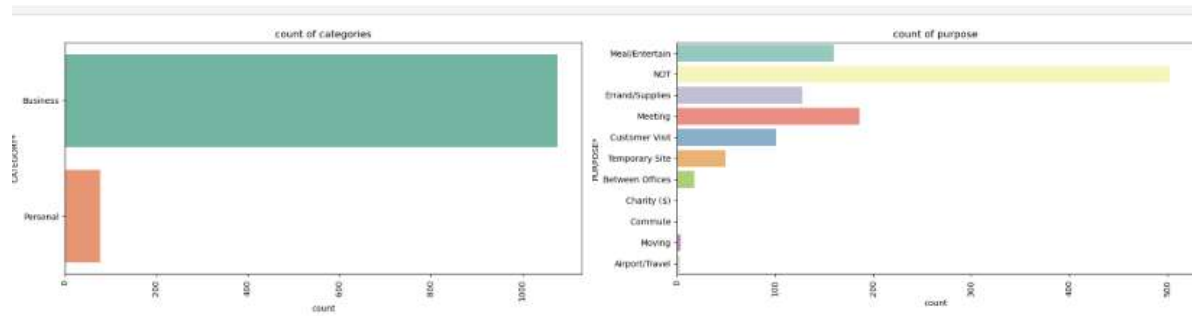
Distribution of Miles



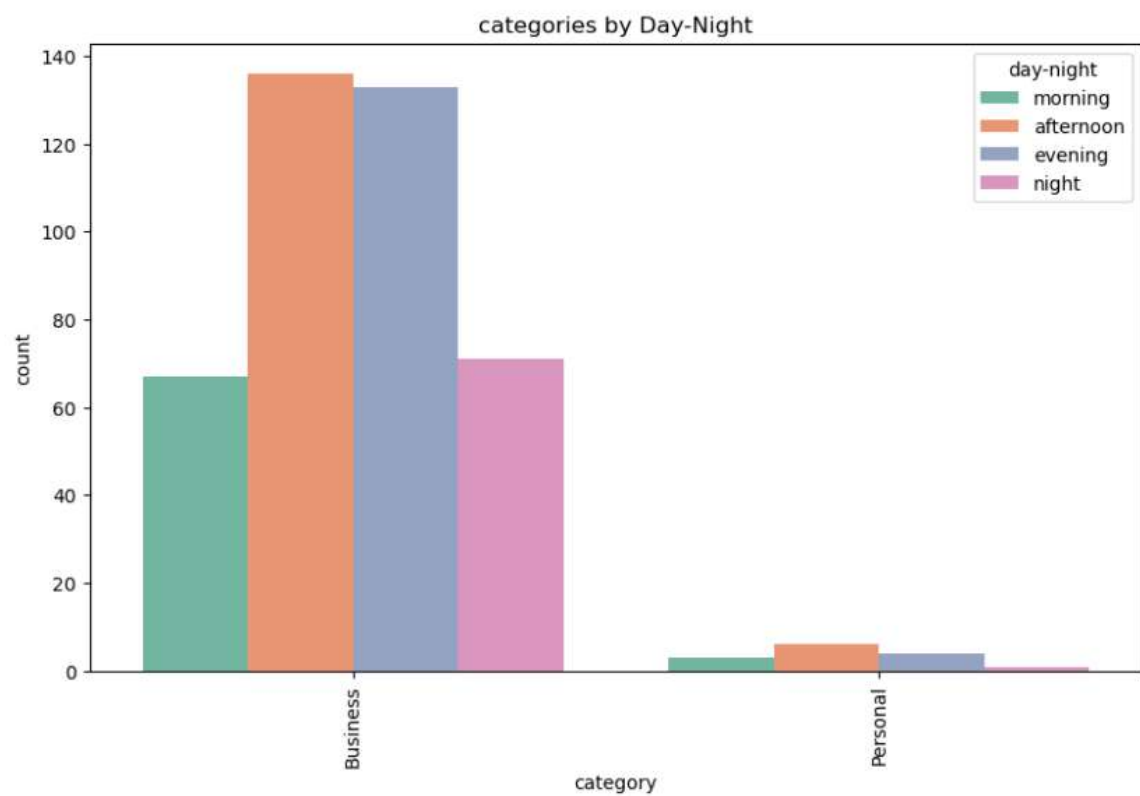
Top 10 Frequent Starting Location



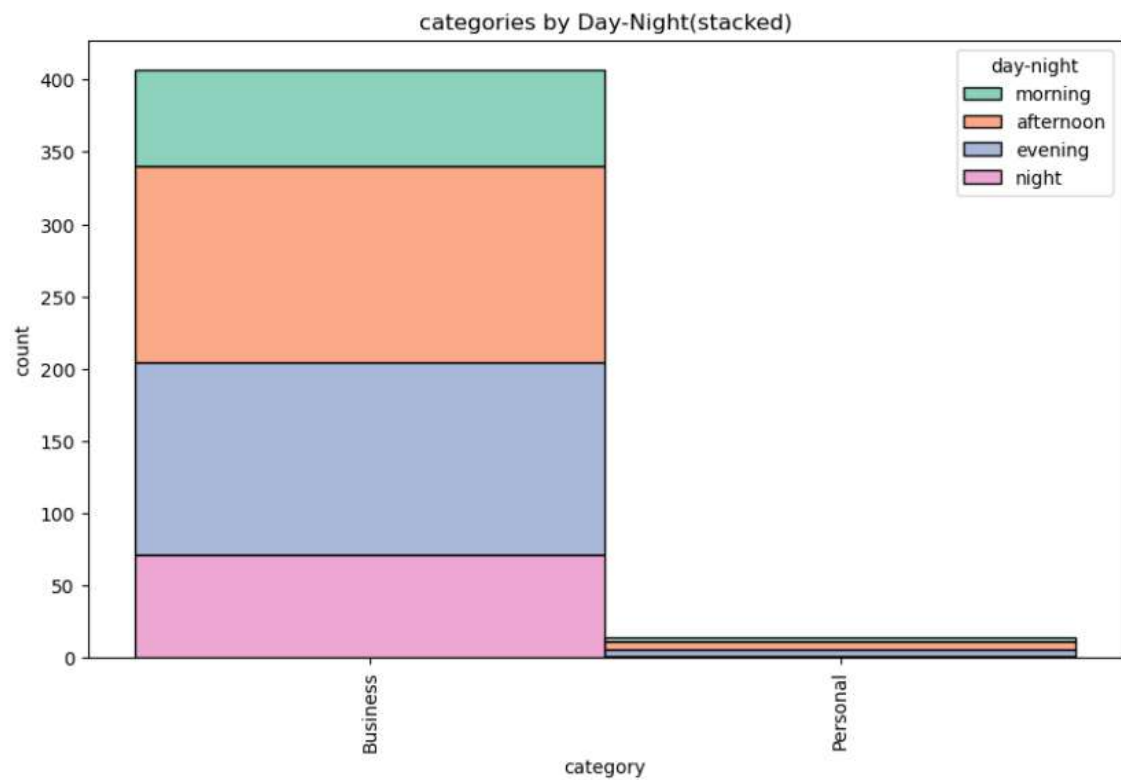
Count of Categories VS Count of Purpose



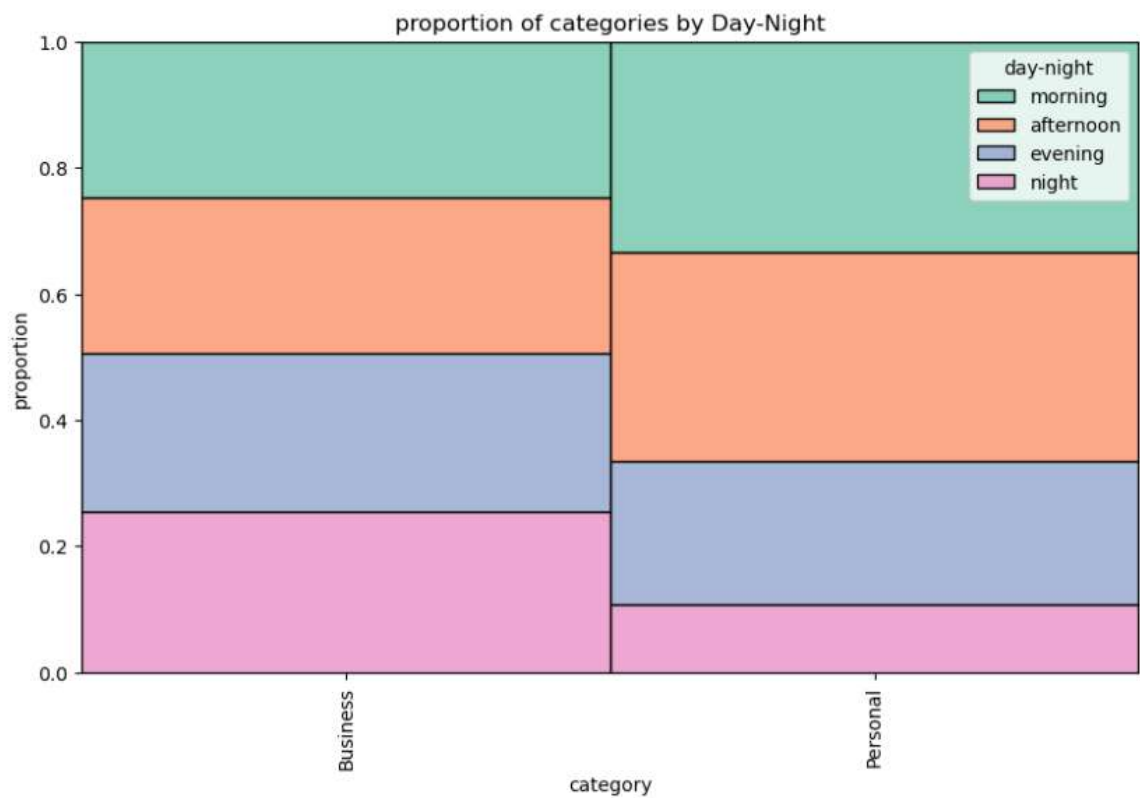
Countplot for Categories By Day-Night



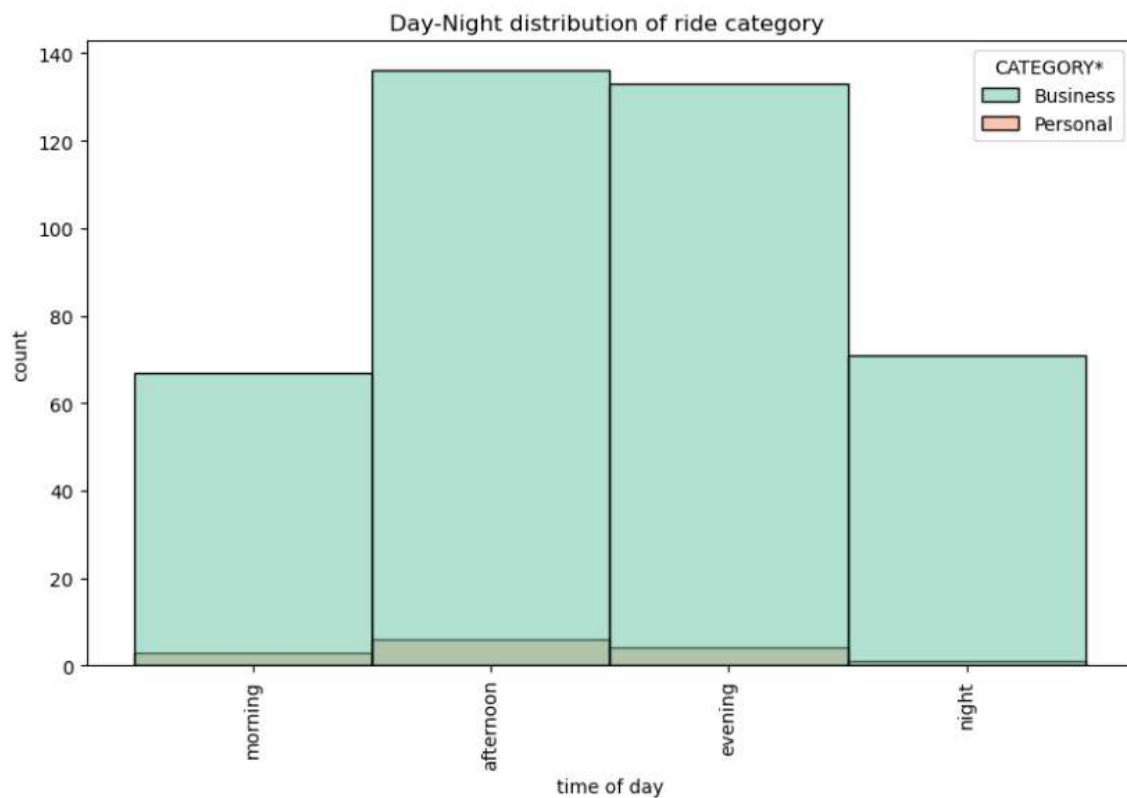
Histplot for Categories By Day-Night (Stacked)



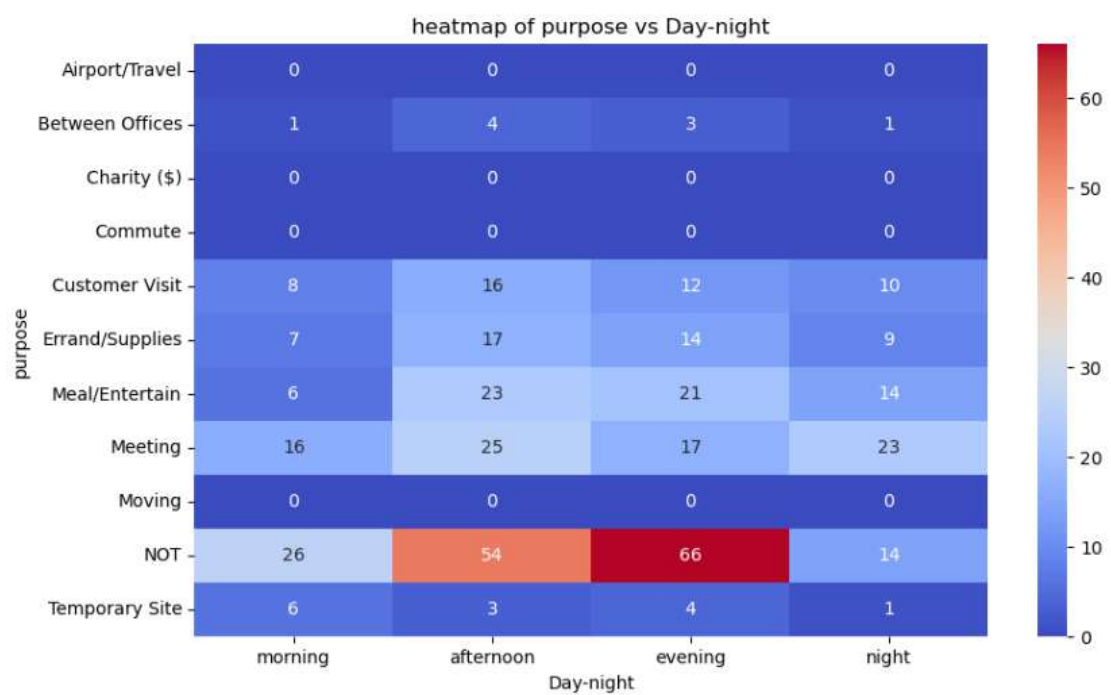
Proportion for Categories By Day-Night



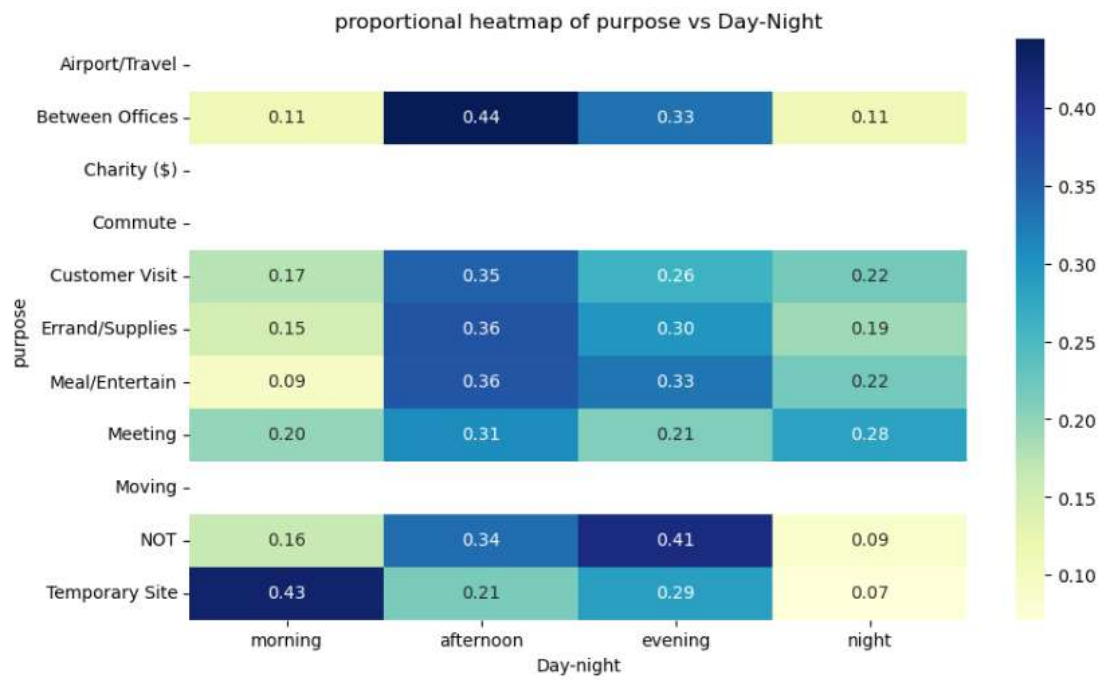
Day-Night Distribution of Ride Category



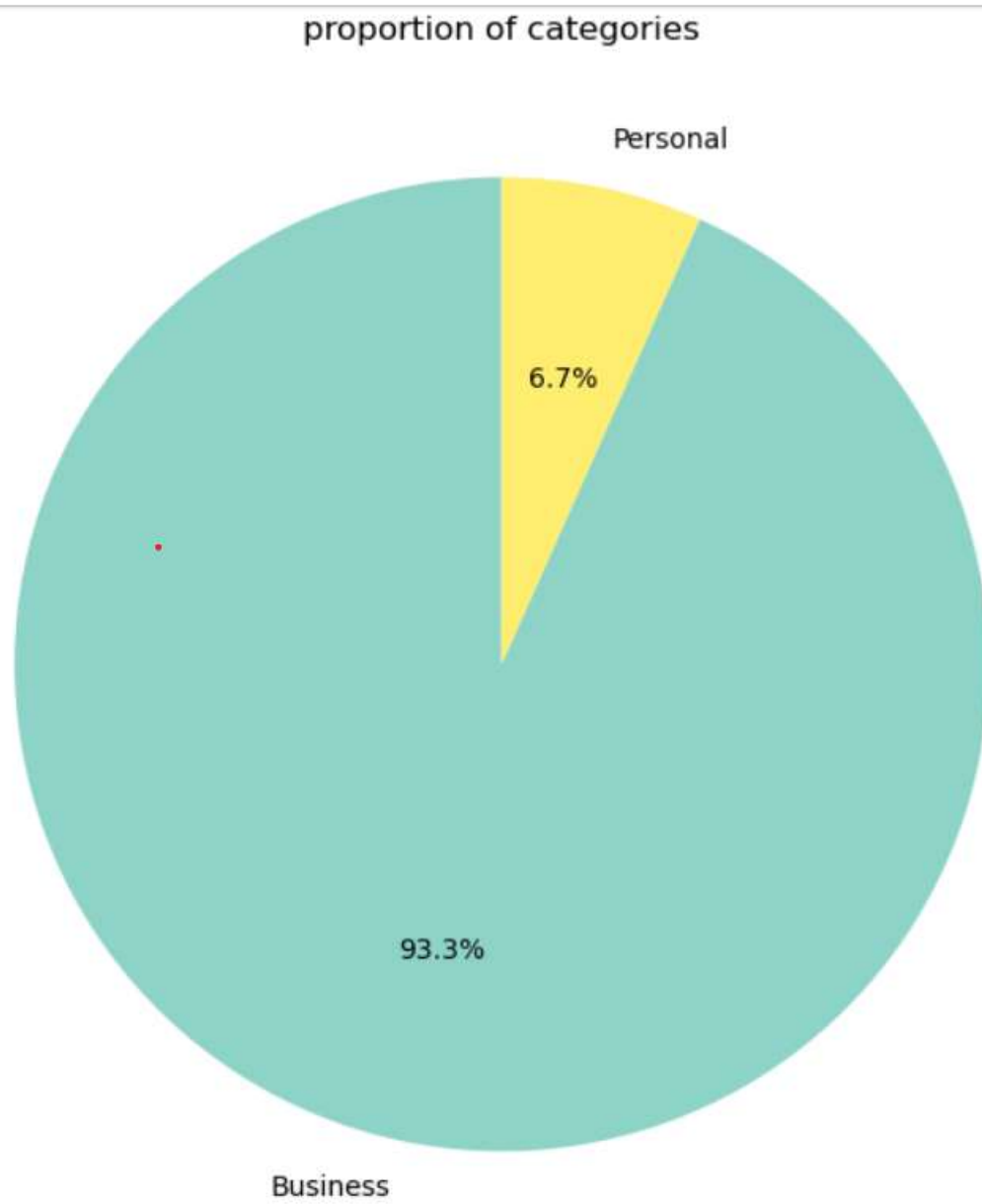
Heatmap of Purpose VS Day-Night



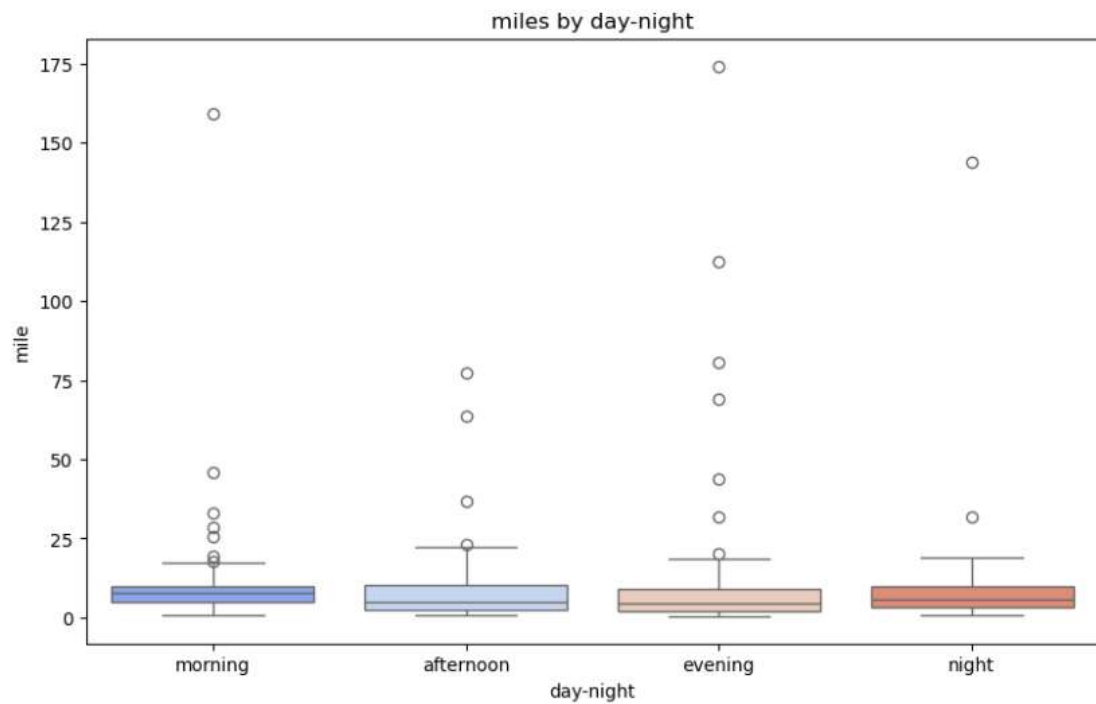
Proportional Heatmap of Purpose VS Day-Night



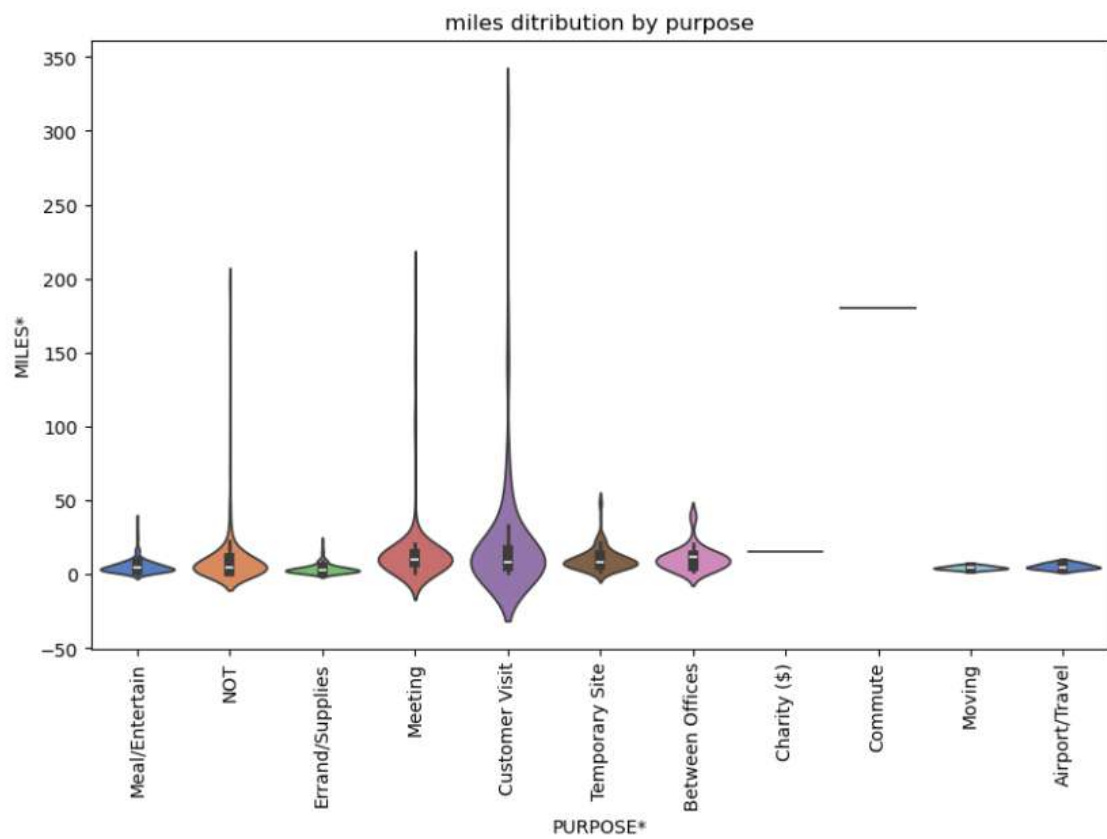
Proportion of Categories



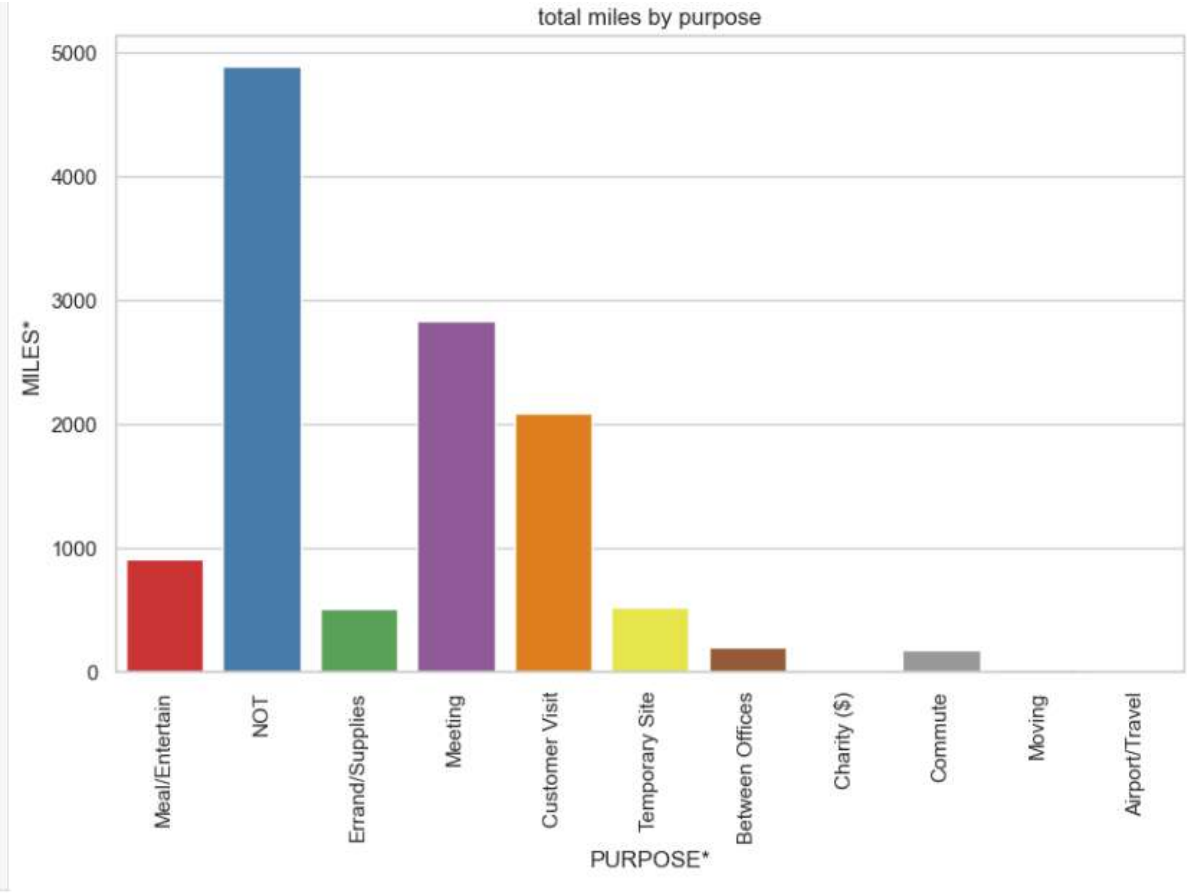
Miles By Day-Night Category



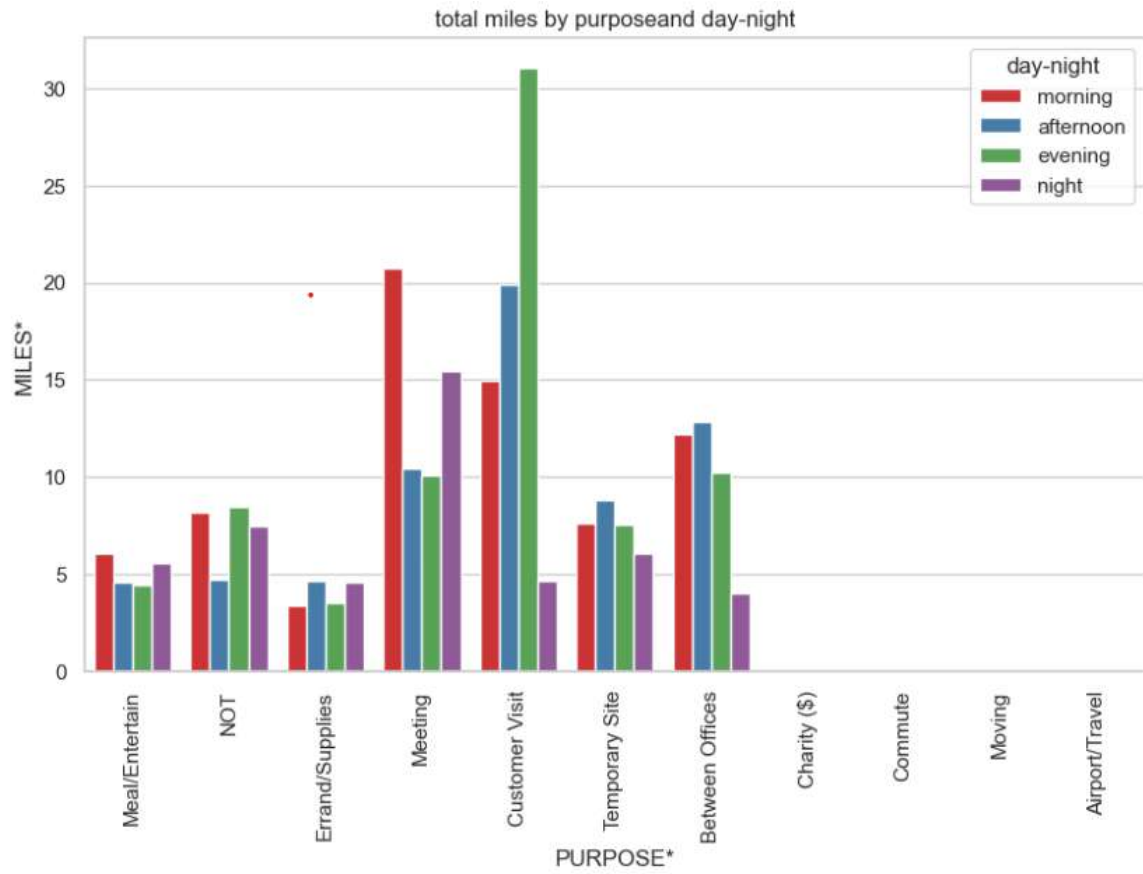
Miles Distribution By Purpose



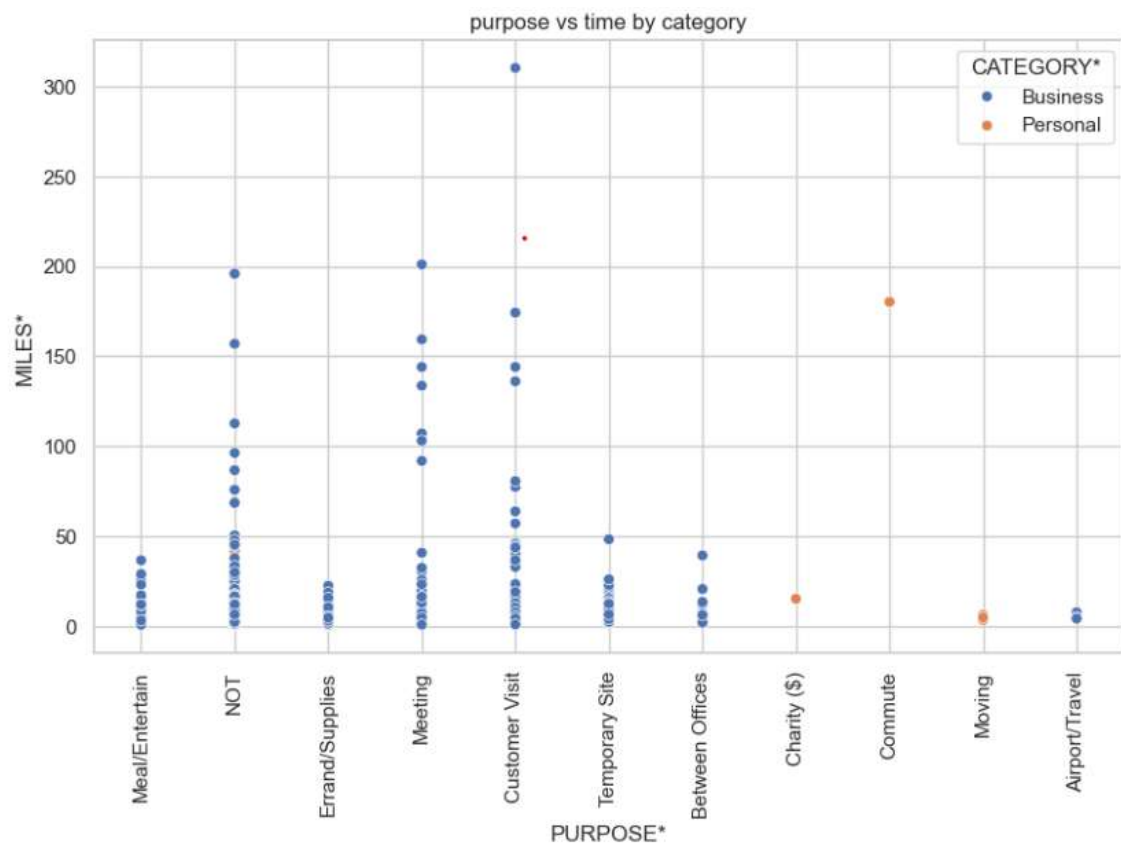
Barplot For Total Miles By Purpose



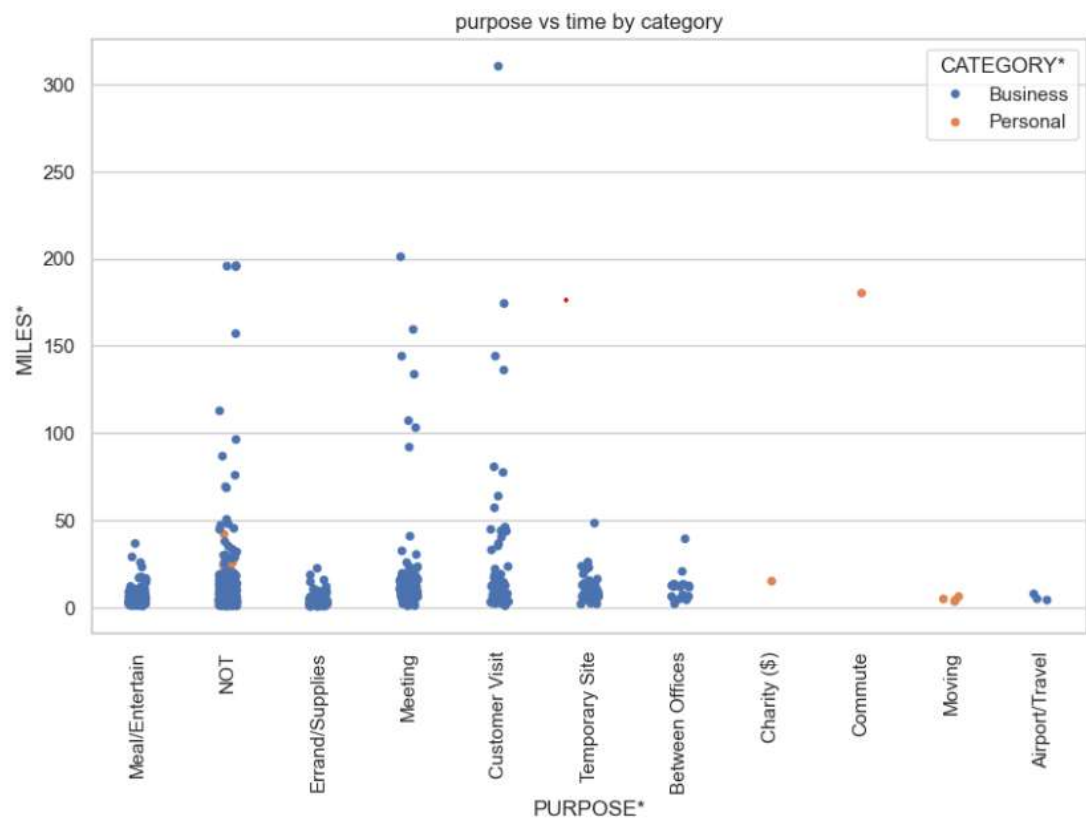
Total Miles By Purpose and Day-Night



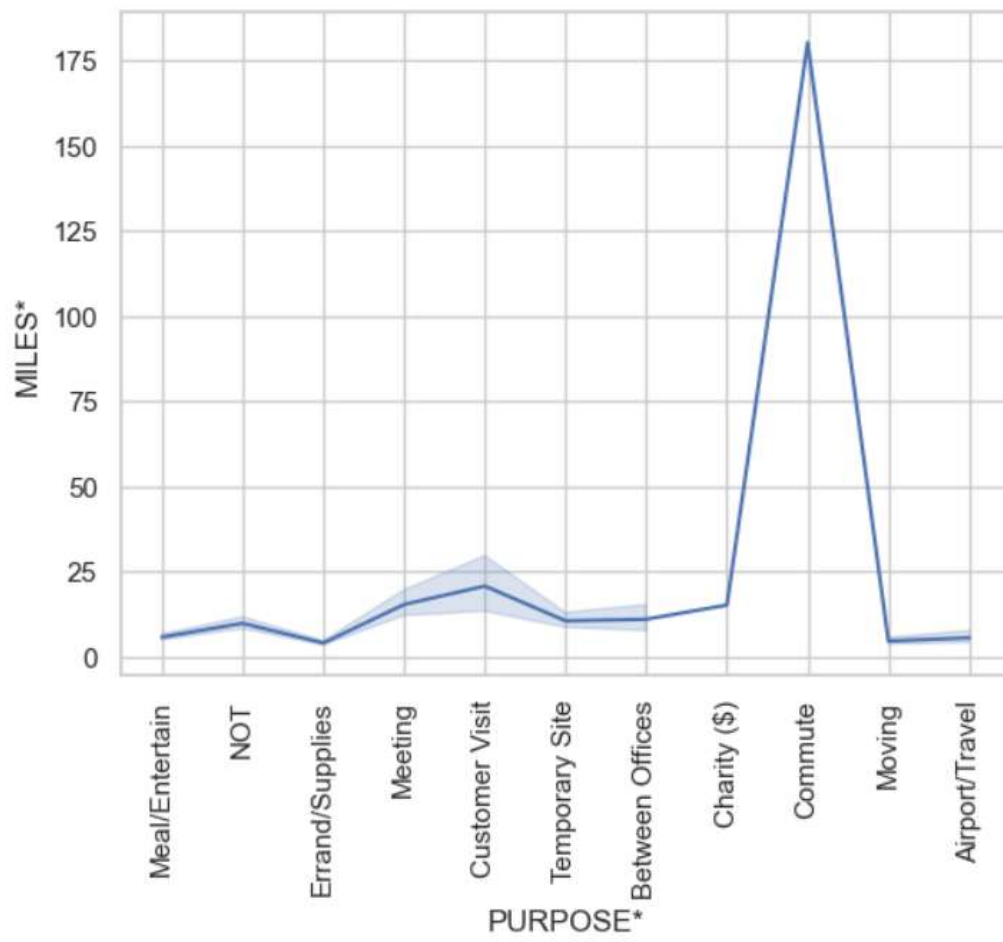
Scatterplot For Purpose VS Time by Category



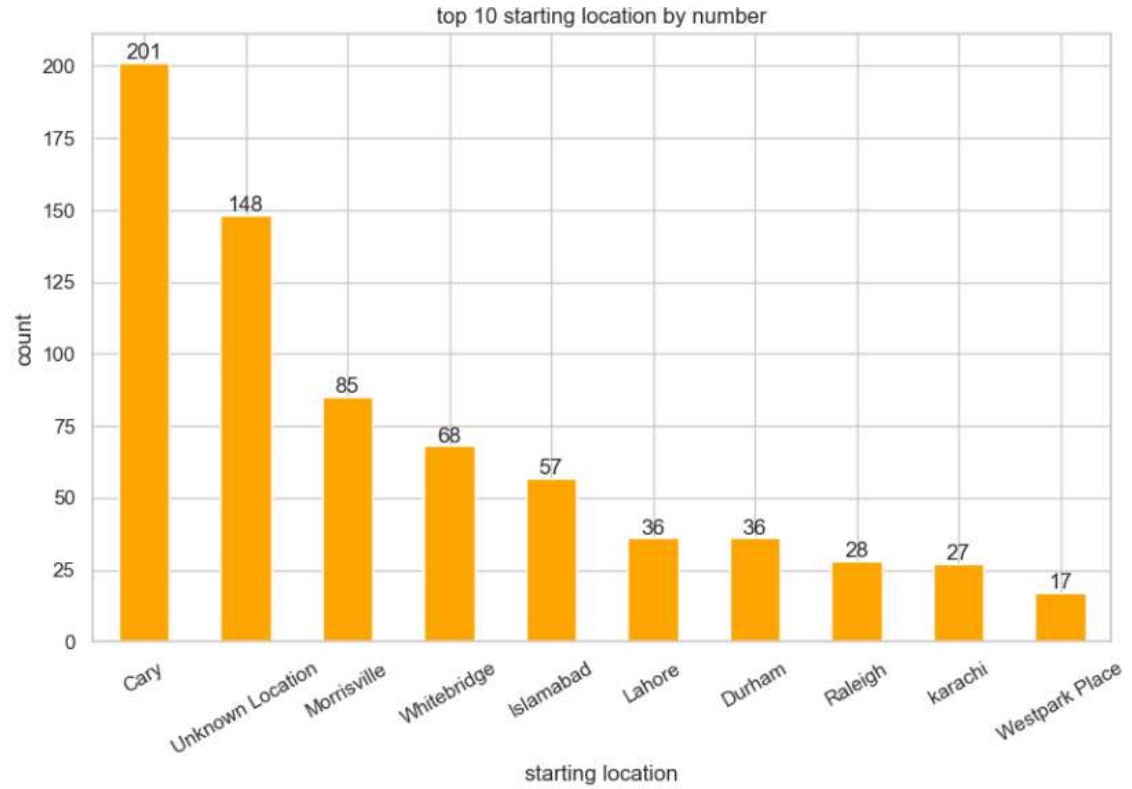
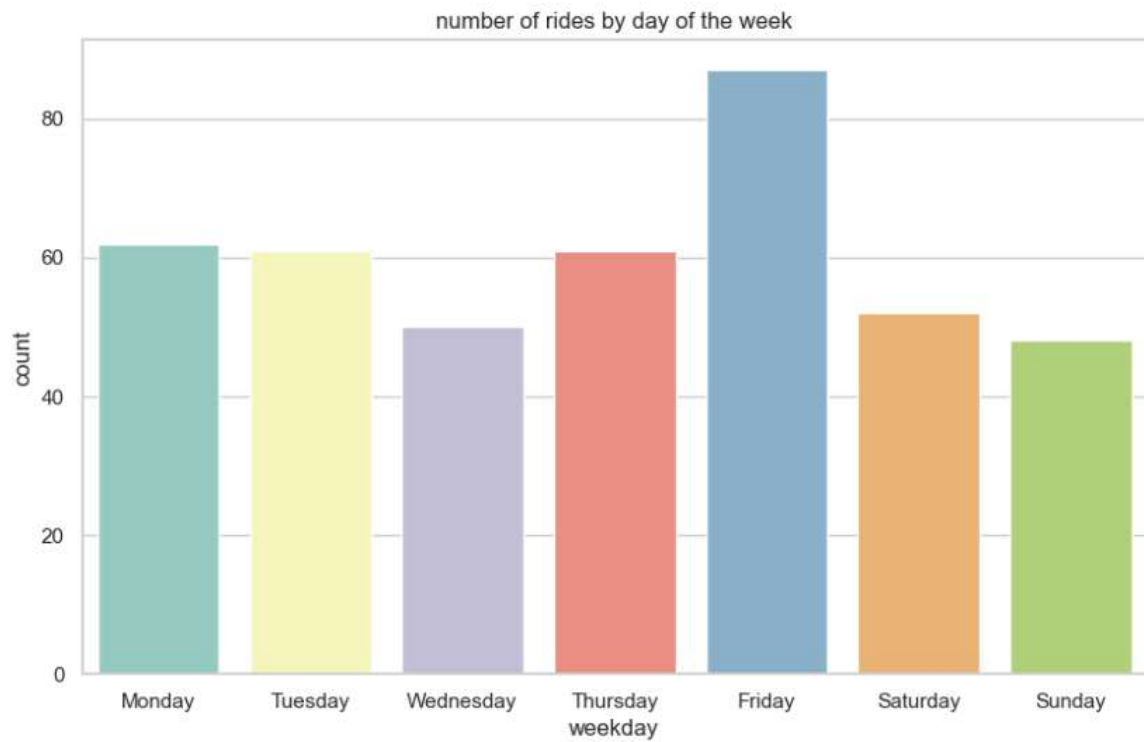
Stripplot For Purpose VS Time By Category



Line Plot For Miles VS Count



Number Of Rides By Day Of The Week



Ride By Time Of Day

